

Section-A

A1: $x(t) = \cos(12\pi t) + \sin(18\pi t)$, $y(t) = x(2t-1)$

Sol: $\Rightarrow y(t) = \cos(12\pi(2t-1)) + \sin(18\pi(2t-1))$
 $= \cos(24\pi t - 12\pi) + \sin(36\pi t - 18\pi) \quad - (1)$

$\therefore$ From trigonometry $\because \cos(\theta - 2\pi n) = \cos \theta$

$\because \sin(\theta - 2\pi n) = \sin \theta$

Applying the trigonometry in eq: (1) we get

$y(t) = \cos(24\pi t) + \sin(36\pi t) \quad - (2)$

$\Rightarrow$ A signal is periodic if there exists T (time period) > 0 i.e. $y(t+T) = y(t)$

Applying $y(t+T) = y(t)$ in the eq: (2) function given

$y(t+T) = \cos(24\pi(t+T)) + \sin(36\pi(t+T))$
 $= \cos(24\pi t + 24\pi T) + \sin(36\pi t + 36\pi T)$

$\Rightarrow$ Cos part

$(24\pi(t+T)) = (24\pi t + 2\pi)$

$\therefore$ Cosine completes one full cycle when the angle increases by 2π .

$\Rightarrow (24\pi t + 24\pi T) = (24\pi t + 2\pi)$

$\Rightarrow 24\pi T = 2\pi$

$\Rightarrow T = \frac{1}{12}$

$\Rightarrow$ Sine part

Sine completes one full cycle $\theta \uparrow$ by 2π

$\Rightarrow \sin(36\pi(t+T)) \quad [36\pi t + 36\pi T = 36\pi t + 2\pi]$

$\Rightarrow 36\pi T = 2\pi$

$T = \frac{1}{18}$

$\Rightarrow$ Now for the total Time period of $y(t)$ least common multiple of $[\frac{1}{12}, \frac{1}{18}]$ is $\frac{1}{6}$

$\therefore$ So the fundamental period of $y(t) = \frac{1}{6}$ s

A2:

Statement: if $x(t)$ is a non zero periodic signal with finite energy over one period then its total energy must be infinite and its average power must be strictly positive

A) True.

$\Rightarrow x(t)$ is periodic with fundamental period T_0

$x(t) = x(t+T_0) \quad - (1)$

$\Rightarrow$ The energy of the $x(t)$

$E_0 = \int_0^{T_0} |x(t)|^2 dt \quad \& \quad E_0 > 0 \quad - (2)$

$\Rightarrow$ Calculating total energy (For a continuous time signal)

$E = \int_{-\infty}^{\infty} |x(t)|^2 dt \Rightarrow E = E_0 + E_0 + \dots \infty$

$\Rightarrow$ Since $E_0 > 0 \quad E = E_0 + E_0 + \dots \infty > 0 \Rightarrow E > 0$
 $\& \quad E = \infty \quad - (3)$

$\Rightarrow$ Average power of $x(t)$

$P = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt \quad - (4)$

$\Rightarrow P = \frac{E_0}{T_0}$

$\Rightarrow P > 0$ since $E_0 > 0 \quad \& \quad T_0 > 0. \quad - (5)$

Example: $x(t) = 2\cos(4\pi t)$

$T_0 = \frac{2\pi}{4\pi} = \frac{1}{2}$

$\Rightarrow E_0 = \int_0^{1/2} |2\cos(4\pi t)|^2 dt$

$= \int_0^{1/2} 4\cos^2(4\pi t) dt$

$= \left[4 \left[\frac{T_0}{2}\right]\right]_0^{1/2} \therefore \int_0^{T_0} \cos^2(t) dt = \frac{T_0}{2}$

$= 4 \left[\frac{1}{4}\right]$

$E_0 = 1$

$\Rightarrow E_{\text{total}} = 1 + 1 + \dots \infty$

$\Rightarrow P = \frac{E_0}{T_0} \Rightarrow \frac{1}{1/2} = 2 > 0$

A3: 01. Average value is zero

Given half wave symmetry

$x(t+T_0/2) = -x(t)$

$\Rightarrow X = \frac{1}{T_0} \int_0^{T_0} x(t) dt$

$= \frac{1}{T_0} \left[\int_0^{T_0/2} x(t) dt + \int_{T_0/2}^{T_0} x(t) dt \right] \quad - (1)$

$\therefore$ Since it is a half wave symmetry

$\int_0^{T_0/2} x(t) dt = - \int_{T_0/2}^{T_0} x(t) dt \quad - (2)$

$$\int_0^{T_0} x(t) dt = 0$$

$$X = 0$$

Statement one is true

Q2. Average value of $|x(t)|$ is zero.

Average the absolute value

$$X_a = \frac{1}{T_0} \int_0^{T_0} |x(t)| dt$$

$$\text{Here } |x(t)| = |-x(t)|$$

$$\begin{aligned} X_a &= \frac{1}{T_0} \left[\int_0^{T_0/2} |x(t)| dt + \int_{T_0/2}^{T_0} |x(t)| dt \right] \\ \int_0^{T_0/2} |x(t)| dt &\neq - \int_{T_0/2}^{T_0} |x(t)| dt \quad - (3) \end{aligned}$$

$$\text{So } X_a \neq 0$$

$$X_a > 0$$

$$\text{Example: } x(t) = \begin{cases} 1 & 0 < t < 1 \\ -1 & 1 < t < 2 \end{cases}$$

$$X = \frac{1}{2} \left[\int_0^1 1 dt + \int_1^2 (-1) dt \right]$$

$$= \frac{1}{2} [1 - 1] = 0$$

$$X_a = \frac{1}{2} \int_0^1 1 dt + \frac{1}{2} \int_1^2 -1 dt$$

$$= \frac{1}{2} [1 + 1] = 1$$

A4

$$\Rightarrow \text{Given } E(x) = E(x_e) + E(x_o)$$

w.k.T

$$x_e(t) = \frac{x(t) + x(-t)}{2} \quad - (1)$$

$$x_o(t) = \frac{x(t) - x(-t)}{2} \quad - (2)$$

$$x_e(-t) = x_e(t) \quad - (3)$$

$$x_o(-t) = -x_o(t) \quad - (4)$$

$\Rightarrow$ Now

$$\langle x_e, x_o \rangle = \int_{-\infty}^{\infty} x_e(t) x_o(t) dt$$

$$\langle x_e, x_o \rangle = \text{even} \times \text{odd} = \text{odd}$$

$\langle x_e, x_o \rangle = 0$ [Integral of an odd funⁿ over symmetric is 0]

$\therefore$ Hence they are orthogonal.

$\Rightarrow$ Energy of $x(t)$

$$E_{x(t)} = \int_{-\infty}^{\infty} |x_e + x_o|^2 dt$$

$$E_{x(t)} = E_{x_e} + E_{x_o} + 2 \langle x_e, x_o \rangle$$

$$\text{we saw } \langle x_e, x_o \rangle = 0$$

$$\therefore E_{x(t)} = E_{x_e} + E_{x_o}$$

AP24110040009

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$$\text{A5: Given } q(t) = r(t+1) - 2r(t) + r(t-1)$$

$$r(t) = t u(t)$$

$$\Rightarrow q(t) \quad t = -1, 0, 1$$

$$\text{1. } t < -1$$

$$r(t+1) = r(t) = r(t-1) = 0$$

$$q(t) = r(t+1) - 2r(t) + r(t-1)$$

$$= 2r(t) - 2r(t)$$

$$= 0$$

$$\text{2. } -1 < t < 0$$

$$r(t+1) = t+1$$

$$r(t) = 0$$

$$r(t-1) = 0$$

$$q(t) = t+1 + 2[0] + 0$$

$$= t+1$$

$$\text{3. } 0 < t < 1$$

$$r(t+1) = t+1$$

$$r(t) = t$$

$$r(t-1) = 0$$

$$q(t) = t+1 - 2(t) + 0$$

$$= -t+1$$

$$\text{4. } t > 1$$

$$q(t) = 0$$

$$\Rightarrow q(t) = \begin{cases} 0 & t < -1 \\ t+1 & -1 < t < 0 \\ -t+1 & 0 < t < 1 \\ 0 & t > 1 \end{cases}$$

First derivative

$$\frac{dq}{dt} = u(t+1) - 2u(t) + u(t-1)$$

Second derivative

$$\Rightarrow \frac{d}{dt} u(t) = \delta(t)$$

$$\Rightarrow \frac{d^2 q}{dt^2} = \delta(t+1) - 2\delta(t) + \delta(t-1)$$

A6: let's take $k(t)$ fundamental period T_1
 $J(t)$ fundamental period T_2

$$D(t) = k(t) + J(t)$$

$\Rightarrow D(t)$ is always periodic? $\rightarrow$ false

The $D(t)$ guaranteed to be periodic if the periods of $k(t)$ & $J(t)$ is commensurate.

$$\frac{T_1}{T_2} \in \mathbb{Q}$$

$$Q1: T_1 = 2 \quad T_2 = 3 \Rightarrow LCM(2, 3) = 6 \checkmark$$

$Q2: T_1 = 1 \quad T_2 = \sqrt{2} \Rightarrow$ arbitrary real periods sum need not be periodic

$\Rightarrow$ Is the fundamental period always LCM? false

$\Rightarrow$ Even when T_1 and T_2 have common period The sum can have a smaller fundamental period than $LCM(T_1, T_2)$

Example:

$$k(t) = \sin(\pi t) + \sin(2\pi t)$$

$$\Rightarrow T_1 = 2 \quad T_2 = 1$$

$k(t)$ time period fundamental = 2

$$J(t) = -\sin(\pi t) + \sin(2\pi t)$$

$\Rightarrow J(t)$ time period fundamental = 2

$$\Rightarrow LCM(T_k, T_J) \Rightarrow LCM(2, 2) = 2$$

Now add

$$k(t) + J(t) = D(t)$$

$$= \sin \pi t + \sin(2\pi t) + [-\sin \pi t] + \sin 2\pi t$$

$$= 2\sin(2\pi t)$$

$$\text{Now } T_D = \frac{2\pi}{2\pi} = 1$$

$$\therefore LCM = 2 \quad \& \quad T_D = 1$$

Section: B

$$B1: x(t) = \begin{cases} -1 & -3 \leq t < -1 \\ 2+t & -1 \leq t < 1 \\ 3 & 1 \leq t < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$y(t) = 1 - 2x(3-2t)$$

i) Sequence

$$\therefore x(3-2t)$$

$$= -2x\left[-\frac{3}{2} + t\right]$$

1 Time reversal $x(t) \Rightarrow x(-t)$

2 Time compression $x(t) \Rightarrow x(-t)$

3 Right shift by $\frac{3}{2}$

$$x(t) \Rightarrow x(-2t) \Rightarrow x\left[-2\left(t - \frac{3}{2}\right)\right]$$

and then

$$\text{multiply by } -2 \Rightarrow -2x\left[-2\left(t - \frac{3}{2}\right)\right]$$

$$\text{shift amplitude by } 1 \Rightarrow 1 - 2x\left[-2\left(t - \frac{3}{2}\right)\right]$$

ii) Transformed breakpoints

$x(t)$ original breakpoints $-3, -1, 1, 2$

$\Rightarrow$ New breakpoints

$$3 - 2t = a$$

$$t = \frac{3-a}{2} \xrightarrow{\text{originals}}$$

Now equate it with originals

$$1 \quad t = \frac{3+3}{2} \Rightarrow t = 3$$

$$2 \quad t = \frac{3+1}{2} \Rightarrow t = 2$$

$$3 \quad t = \frac{3-1}{2} \Rightarrow t = 1$$

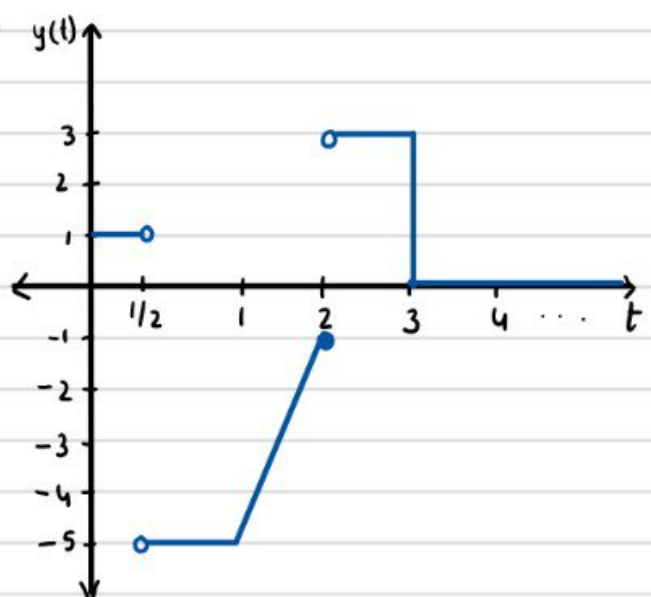
$$4 \quad t = \frac{3-2}{2} \Rightarrow t = \frac{1}{2}$$

new breakpoints $0, \frac{1}{2}, 1, 2, 3$

iii) $y(t)$

$$y(t) = \begin{cases} 1 & ; t < \frac{1}{2} \\ -5 & ; \frac{1}{2} < t < 1 \\ 4t-9 & ; 1 \leq t \leq 2 \\ 3 & ; 2 < t \leq 3 \\ 1 & ; t > 3 \end{cases}$$

iv)



V. Take $x(3-2t)$

$$Xx(t) \rightarrow x(t-3) \rightarrow x(2t-3)$$

$$\checkmark x(-2) \left(-\frac{3}{2} + t\right)$$

$$B2: x(t) = \begin{cases} -1 & -3 \leq t < -1 \\ t+2 & -1 \leq t < 1 \\ 3 & 1 \leq t < 2 \\ 0 & \text{otherwise} \end{cases}$$

for any signal

$$x_e(t) = \frac{x(t) + x(-t)}{2}; \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

1 find $x(-t)$

$$\text{O1 } x(t) = -1 \quad -3 \leq t < -1$$

$$\Rightarrow t \rightarrow -t \text{ so } 3 \geq t > 1$$

$$x(-t) = -1; \quad 3 \geq t > 1$$

$$\text{O2 } x(t) = 2+t; \quad -1 \leq t < 1$$

$$\Rightarrow t \rightarrow -t \text{ so } -1 < t \leq 1$$

$$x(-t) = 2-t; \quad -1 < t \leq 1$$

$$\text{O3 } x(t) = 3 \quad 1 \leq t < 2$$

$$t \rightarrow -t \text{ so } 1 \leq -t < 2$$

$$x(-t) = 3; \quad -2 < t \leq 1$$

$$x(-t) = \begin{cases} 3 & -2 < t < -1 \\ 2-t & -1 < t < 1 \\ -1 & 1 < t < 3 \\ 0 & \text{otherwise} \end{cases}$$

$\Rightarrow x_e(t)$

1 for $-3 < t < -2$

$$x_e(t) = \frac{-1+0}{2} = -\frac{1}{2}$$

2 for $-2 < t < -1$

$$x_e(t) = \frac{-1+3}{2} = 1$$

3 for $-1 < t < 1$

$$x_e(t) = \frac{2+t+2-t}{2} = 2$$

4 $1 < t < 2$

$$x_e(t) = \frac{3+(-1)}{2} = 1$$

5 $2 < t < 3$

$$x_e(t) = \frac{0+(-1)}{2} = -\frac{1}{2}$$

$$x_e(t) = \begin{cases} 0 & |t| > 3 \\ -\frac{1}{2} & -3 < t < -2 \\ 1 & -2 < t < -1 \\ 2 & -1 < t < 1 \\ 1 & 1 < t < 2 \\ -\frac{1}{2} & 2 < t < 3 \end{cases}$$

$\Rightarrow x_o(t)$

1 for $-3 < t < -2$

$$x_o(t) = \frac{-1-0}{2} = -\frac{1}{2}$$

2 for $-2 < t < -1$

$$x_o(t) = \frac{-1-3}{2} = -2$$

3 for $-1 < t < 1$

$$x_o(t) = \frac{2+t-(2-t)}{2} = t$$

4 $1 < t < 2$

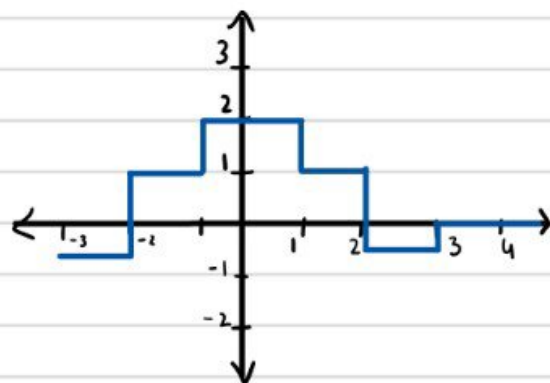
$$x_o(t) = \frac{3-(-1)}{2} = 2$$

5 $2 < t < 3$

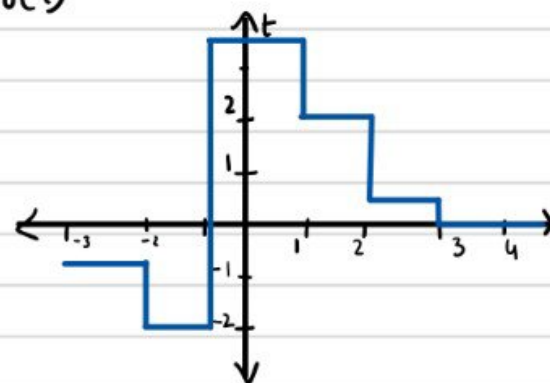
$$x_o(t) = \frac{0-(-1)}{2} = \frac{1}{2}$$

$$x_o(t) = \begin{cases} 0 & |t| > 3 \\ -\frac{1}{2} & -3 < t < -2 \\ -2 & -2 < t < -1 \\ t & -1 < t < 1 \\ 2 & 1 < t < 2 \\ \frac{1}{2} & 2 < t < 3 \end{cases}$$

$x_e(t)$



$x_o(t)$



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B3: Given $x(t) = 2\text{rect}\left(\frac{t-1}{2}\right) - \text{rect}\left(\frac{t+2}{4}\right)$

$y(t) = -3x\left(-\frac{t+1}{2}\right) + 2$

lets say $k = -\frac{t-1}{2}$

$\Rightarrow x(t) = 2\text{rect}\left(\frac{t-1}{2}\right) - \text{rect}\left(\frac{t+2}{4}\right)$

$\hat{=} 2\text{rect}\left(\frac{t-1}{2}\right)$

$-\frac{1}{2} < \frac{t-1}{2} < \frac{1}{2}$

$\Rightarrow -1 < t-1 < 1$

$\Rightarrow 0 < t < 2$

So height = 2 ; $0 < t < 2$

$\hat{=} -\text{rect}\left(\frac{t+2}{4}\right)$

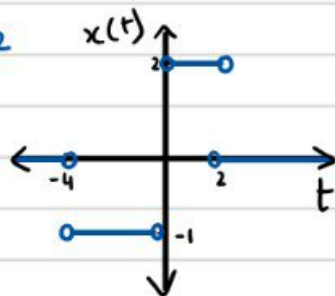
$\Rightarrow -\frac{1}{2} < \frac{t+2}{4} < \frac{1}{2}$

$\Rightarrow -2 < t+2 < 2$

$\Rightarrow -4 < t < 0$

height = -1 ; $-4 < t < 0$

$\hat{=} \text{elsewhere } 0$



$\Rightarrow \text{Now } y(t) = -3x\left(-\frac{t-1}{2}\right) + 2$

$\hat{=} \text{original } x(t)$

2 ; $0 < t < 2$

-1 ; $-4 < t < 0$

$\hat{=} \text{Now } x\left(-\frac{t-1}{2}\right)$

$\Rightarrow -\frac{t-1}{2} = u$

$\Rightarrow x(u)$

i. $0 < u < 2$

$0 < -\frac{t-1}{2} < 2$

$0 < -t-1 < 4$

$1 < -t < 5$

$-1 > t > -5$

$-5 < t < -1$

ii. $-4 < u < 0$

$-4 < -\frac{t-1}{2} < 0$

$-8 < -t-1 < 0$

$-7 < -t < 1$

$7 > t > -1$

$-1 < t < 7$

$\Rightarrow \text{Heights}$

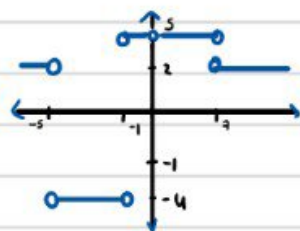
$-3x(u)$

$\Rightarrow \text{i. } +2 \times (-3) = -6$

$\text{ii. } -1 \times (-3) = +3$

$\Rightarrow +2 \rightarrow -6$

$-1 \rightarrow +3$



$\therefore$ The u changes the width, location while -3 and $+2$ change heights

B4: Given

$\Rightarrow x(-t) = -x(t)$

$x(t+u) = x(t)$

$x(t) = 0$ for $1 < |t| < 2$

$\Rightarrow \hat{=} \text{odd signal from } x(-t) = -x(t)$
 $\hat{=} T = 4$ from $x(t+u) = x(t)$

$\Rightarrow$ lets take usec time limits $-2 \leq t < 2$

$\hat{=} x(t) = 0$ $1 < |t| < 2$

So $x(t) = 0$ $\begin{cases} (-2, -1) \\ (1, 2) \end{cases}$

$\Rightarrow x(t) = \begin{cases} -1 ; -1 < t < 0 \\ 0 ; t = 0 \\ 1 ; 0 < t < 1 \\ 0 ; 1 \leq t < 2 \end{cases}$

$\Rightarrow \int_{-2}^2 x(t) dt = 0$

Since it is a odd signal

$\Rightarrow$ Zero average doesn't mean RMS zero

$\Rightarrow x_{RMS} = \sqrt{\frac{1}{4} \left[\int_{-1}^0 1^2 dt + \int_0^1 1^2 dt \right]}$
 $= \sqrt{\frac{1}{4} [2]}$

$x_{RMS} = \frac{1}{\sqrt{2}} \neq 0.$

Section C

C1: Given

a) $K_1(t) = \cos(8t) + \sin(12t)$

$T_1 = \frac{2\pi}{8} = \frac{\pi}{4}$

$T_2 = \frac{2\pi}{12} = \frac{\pi}{6}$

Now frequency ratio = $\frac{2}{3}$ (rational so periodic)

$\omega_{\text{smallest}} = \text{gcd}(8, 12) = 4$

So $T_0 = \frac{2\pi}{\omega_0}$

$T_0 = \frac{2\pi}{4} = \frac{\pi}{2}$

$$b) k_2(t) = \cos(2\pi t) + \sin(2\pi \sqrt{2} t)$$

$$T_1 = 1 \quad T_2 = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi\sqrt{2}}$$

$$T_2 = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{1}{1/\sqrt{2}} = \sqrt{2} = \text{irrational not periodic}$$

$$c) k_3(t) = \cos^2(5t) + \sin^2(7t)$$

$$\stackrel{!}{=} \cos^2(5t) \Rightarrow \frac{1 + \cos(10t)}{2} = \frac{1}{2} + \frac{1}{2}\cos(10t)$$

$$\stackrel{!}{=} \sin^2(7t) \Rightarrow \frac{1 - \cos(14t)}{2} = \frac{1}{2} - \frac{1}{2}\cos(14t)$$

$$\text{now } k_3(t) = 1 + \frac{1}{2}\cos(10t) - \frac{1}{2}\cos(14t)$$

$$\Rightarrow \text{Now } \frac{10}{14} = \frac{5}{7} \text{ so rational } \Rightarrow \text{periodic}$$

$$\& \text{ fundamental } \gcd(10, 14) = 2$$

$$T_0 = \frac{2\pi}{2}$$

$$T_0 = \pi$$

$$d) k_4(t) = \cos(4\pi t) + \cos(6\pi t) + \cos(10\pi t)$$

$$\gcd(4\pi, 6\pi, 10\pi) = 2\pi$$

$$T_0 = \frac{2\pi}{2\pi} = 1$$

C2: Given

$$x(t) = \begin{cases} 2t/1ms & ; 0 \leq t < 1ms \\ 2 & ; 1ms \leq t < 2ms \\ \frac{2(3ms-t)}{1ms} & ; 2ms \leq t < 3ms \end{cases}$$

$$T_0 = 6ms \quad 0 \leq t < 3ms$$

$$x(t+3ms) = -x(t)$$

$$\stackrel{01}{=} X = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

Since it is half wave symmetry

$$\int_3^6 x(t) dt = - \int_0^3 x(t) dt$$

$$\Rightarrow \int_0^6 x(t) dt = 0$$

$$x_{avg} = 0$$

$$\stackrel{02}{=} |X| = \frac{1}{T_0} \int_0^{T_0} |x(t)| dt$$

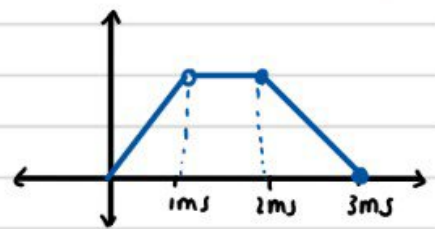
$$\Rightarrow |X| = \frac{1}{6} \int_0^6 |x(t)| dt$$

$$= \frac{2}{6} \int_0^3 x(t) dt$$

$$= \frac{2}{6} [4]$$

$$|X| = \frac{4}{3}$$

$$[\because \int_0^3 x(t) dt = 4]$$



03 RMS Value.

$$X_{rms} = \sqrt{\frac{1}{T_0} \int_0^{T_0} x^2(t) dt}$$

$$x(t+3) = -x(t)$$

$$x^2(t+3) = x^2(t)$$

$$x_{rms}^2 = \frac{2}{6} \int_0^3 x^2(t) dt$$

$$\stackrel{!}{=} \Rightarrow x(t) = 2t \quad 0 \leq t < 1$$

$$\Rightarrow x^2(t) = 4t^2$$

$$\Rightarrow \int_0^1 4t^2 dt = \frac{4}{3}$$

$$\stackrel{2}{=} \Rightarrow x(t) = 2 \quad 1 \leq t < 2$$

$$\Rightarrow \int_1^2 x^2(t) dt$$

$$\Rightarrow \int_1^2 4 dt = 4$$

$$\stackrel{3}{=} \Rightarrow x(t) = 2(3-t) \quad 2 \leq t < 3$$

$$\Rightarrow x^2(t) = 4(3-t)^2$$

$$\int_2^3 4(3-t)^2 dt = \frac{4}{3}$$

$$\Rightarrow \frac{20}{3}$$

$$x_{rms}^2 = \frac{20}{9}$$

$$x_{rms} = \frac{\sqrt{20}}{3}$$

$$\stackrel{4}{=} P_{avg} = x_{rms}^2$$

$$P_{avg} = \frac{20}{9} //$$

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C3: Given

$$P_x = 9$$

$$y(t) = 2x(t) + 5$$

Q1 Average power of a signal is given by

$$P_x = x_{rms}^2 \quad \& \text{ Given } P_x = 9$$

$$\text{So } 9 = x_{rms}^2$$

$$\Rightarrow x_{rms} = 3$$

$$Q2 \quad y(t) = 2x(t) + 5$$

$$P_y = \langle y^2(t) \rangle$$

$$= \langle (2x(t) + 5)^2 \rangle$$

$$= \langle 4x^2(t) + 20x(t) + 25 \rangle$$

$$= 4 \langle x^2(t) \rangle + 20 \langle x(t) \rangle + 25$$

$$\text{w.k.T } \langle x^2(t) \rangle = P_x$$

$$= 4P_x + 20 \langle x(t) \rangle + 25$$

Q3 Since the signal is half wave symm,
 $x(t + \frac{T}{2}) = -x(t)$

$\Rightarrow \langle x(t) \rangle$ for a half wave symmetry is zero

$$\langle x(t) \rangle = 0$$

$$\text{So } P_y = 4P_x + 20(0) + 25$$

$$= 36 + 25$$

$$\Rightarrow P_y = 61$$

$$Q4 \quad y_{rms}^2 = P_y$$

$$y_{rms} = \sqrt{P_y}$$

$$y_{rms} = \sqrt{61}$$

Q5 why $(2\sqrt{P_x} + 5)^2$ wrong

Sol: RMS of a sum cannot generally be obtained by adding individual RMS values.

$$\text{correct: } P_y = 4P_x + 20\sqrt{P_x} + 25 = 61$$

$$\text{wrong: } P_y = (2\sqrt{P_x} + 5)^2 = 121$$

Section D

$$Q1. a) \quad k_1(t) = e^{-2|t|}$$

$$E_1 = \int_{-\infty}^{\infty} (k_1(t))^2 dt$$

$$E_1 = \int_{-\infty}^{\infty} e^{-4|t|} dt$$

$$= 2 \int_0^{\infty} e^{-4|t|} dt$$

$\therefore$ Since

it is an even function.

$$= 2 \left[-\frac{1}{4} e^{-4t} \right]_0^{\infty}$$

$$= -\frac{2}{4} \left[e^{-4(\infty)} - e^{-4(0)} \right]$$

$$= -\frac{2}{4} \left[0 - \frac{1}{1} \right]$$

$$= \frac{1}{2}$$

$\therefore$ It is an energy signal the power is zero

$$b) \quad k_2(t) = e^{-t}u(t) + e^t u(-t)$$

$$= e^{-|t|}$$

$$E_2 = \int_{-\infty}^{\infty} e^{-2|t|} dt$$

$$= 2 \int_0^{\infty} e^{-2|t|} dt$$

$\therefore$ even function

$$= 2 \left[-\frac{1}{2} \right]$$

$$E_2 = 1$$

$\therefore$ It is an energy signal

$$c) \quad x_3(t) = 3 + 2\cos(4t)$$

This is a periodic signal so a power signal.

$$P_3 = \langle (3 + 2\cos 4t)^2 \rangle$$

$$= \langle 9 + 12\cos 4t + 4\cos^2 4t \rangle$$

$$= 9 + 12 \langle \cos 4t \rangle + 4 \langle \cos^2 4t \rangle$$

$$\text{w.k.T for one period } \langle \cos 4t \rangle = 0 \quad \& \langle \cos^2 4t \rangle = \frac{1}{2}$$

$$= 9 + 0 + 4 \left[\frac{1}{2} \right]$$

$$P_3 = 11 \quad \therefore \text{Power signal.}$$

$$d) \quad x_4(t) = tu(t)$$

$$x_4 = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$E_4 = \int_0^{\infty} t^2 (u(t))^2 dt$$

$\hookrightarrow$ equals to one "unit signal"

$$= \int_0^{\infty} t^2 dt$$

$$= \left[\frac{t^3}{3} \right]_0^{\infty}$$

$$E_4 = \infty$$

$$\text{Now } P_4 = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T (tu(t))^2 dt$$

$$P_4 = \infty$$

$\therefore$ Neither Power nor energy signal

$$e) \quad x_5 = \frac{\sin t}{t}$$

$$E_5 = \int_{-\infty}^{\infty} \left(\frac{\sin t}{t} \right)^2 dt$$

by formula $\int_{-\infty}^{\infty} \left(\frac{\sin t}{t}\right)^2 dt = \pi$

$E_s = \pi$ ∴ Energy signal

D2: Given

$\phi_1(t) = \text{rect}(t/2)$

$\phi_2(t) = \text{sgn}(t) \text{rect}(t/2)$

$\phi_3(t) = (t^2 - \frac{1}{3}) \text{rect}(t/2)$

$q(t) = 3\phi_1(t) - 4\phi_2(t) + 2\phi_3(t)$

w.k.T $\langle \phi_i, \phi_j \rangle = \int_{-\infty}^{\infty} \phi_i(t) \phi_j(t) dt$

or w.k.T

$\text{rect}\left[\frac{t}{T}\right] = \begin{cases} 1; & |t| < \frac{T}{2} \\ 0; & |t| > \frac{T}{2} \end{cases}$

In the function $T=2$

so $|t| < \frac{2}{2} = 1$

$-1 < t < 1$

and the $\text{rect}(\frac{t}{2})$ is zero outside

$\text{rect}\left(\frac{t}{2}\right) = \begin{cases} 1; & -1 < t < 1 \\ 0; & \text{otherwise} \end{cases}$

∴ so for $-1 < t < 1$ $\text{rect}(\frac{t}{2}) = 1$

So for $-1 < t < 1$ here

$\therefore \langle \phi_1, \phi_2 \rangle = \int_{-\infty}^{\infty} \phi_1 \phi_2 dt$

$= \int_{-\infty}^{\infty} (1) [\text{sgn}(t)] dt$

$= 0$ ∴ $\text{sgn}(t) = 0$ it is odd & we have symmetric limits.

or $\langle \phi_2, \phi_3 \rangle$

$\phi_2(t) = \text{sgn}(t) = \text{odd}$

$\phi_3(t) = \frac{t^2}{2} - \frac{1}{3}$ is even

odd x even = odd

∴ odd + symmetric = 0

or $\langle \phi_3, \phi_1 \rangle$

$= \int_{-1}^1 \left(\frac{t^2}{2} - \frac{1}{3}\right) dt$

$= \left[\frac{t^3}{6} - \frac{t}{3}\right]_{-1}^1 = -\frac{1}{3}$

so ϕ_1 & ϕ_3 are not orthogonal to each other. AP24110040009
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$q(t) = 3\phi_1(t) - 4\phi_2(t) + 2\phi_3(t)$

$E_1 = \langle \phi_1, \phi_1 \rangle = \int_{-1}^1 1 dt = 2$

$E_2 = \langle \phi_2, \phi_2 \rangle = \int_{-1}^1 \text{sgn}^2(t) dt$
 $= \int_{-1}^1 1 dt$

$E_2 = 2$

$E_3 = \int_{-1}^1 \left[\frac{t^2}{2} - \frac{1}{3}\right]^2 dt$

$= 2 \int_0^1 \left[\frac{t^2}{2} - \frac{1}{3}\right]^2 dt$

$= 2 \left[\frac{1}{20} - \frac{1}{9} + \frac{1}{9}\right]$

$E_3 = \frac{1}{10}$

Using orthogonality

$E_q = 9E_1 + 16E_2 + 4E_3 + 2[3][2]\langle \phi_1, \phi_3 \rangle$

$= 9(2) + 16(2) + 4\left(\frac{1}{10}\right) + 12\left(-\frac{1}{3}\right)$

$= 18 + 32 + 0.4 + (-4)$

$E_q = 46.4$

D3: Given $x_a(t) = e^{-a|t|}$; $a > 0$

or $\lim_{a \rightarrow 0^+} e^{-a|t|}$

w.k.t $a \rightarrow 0^+$; $-a|t| \rightarrow 0$

$= \lim_{a \rightarrow 0^+} e^{-a|t|} \Rightarrow e^0 = 1$

or $E_{x_a} = \int_{-\infty}^{\infty} (x_a(t))^2 dt$

$x_a(t) = e^{-a|t|} \Rightarrow F_{x_a} = \int_{-\infty}^{\infty} e^{-2a|t|} dt$

Since it is an even function

$F_{x_a} = 2 \int_0^{\infty} e^{-2at} dt$

$E_{x_a} = 2 \left[\frac{1}{2a}\right] \therefore \int_0^{\infty} e^{-kt} dt = \frac{1}{k}$

$E_{x_a} = \frac{1}{a}$

∴ $x_a(t)$ is an energy signal for every value of a i.e. $a > 0$.

$$3. P_{x_a} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x_a(t)|^2 dt$$

now

$$P_{x_a} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-2a|t|} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\int_{-\infty}^{\infty} e^{-2a|t|} dt \right] \quad \text{|| } a$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2Ta}$$

$$= 0$$

$$04 \ x(t)=1$$

$$E_x = \int_{-\infty}^{\infty} 1^2 dt = \infty$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 1 dt = 1$$

D4: Given.

$$E_x = 4 \quad E_y = 9 \quad \langle x, y \rangle = -1$$

$$E_{x+ay} = E_x + a^2 E_y + 2a \langle x, y \rangle$$

$$\stackrel{1}{=} E_{x+y} = E_x + E_y + 2 \langle x, y \rangle$$

$$= 11$$

$$\stackrel{2}{=} E_{2x-y} = 4E_x + E_y - 4 \langle x, y \rangle$$

$$= 4(4) + 9 - 4(-1) = 29$$

$$\stackrel{3}{=} E_{x-2y} = E_x + 4E_y - 4 \langle x, y \rangle$$

$$= 44$$

$$\stackrel{4}{=} \langle x+\alpha y, y \rangle = 0$$

$$\langle a+b, c \rangle = \langle a, c \rangle + \langle b, c \rangle$$

$$\langle x, y \rangle + \alpha \langle y, y \rangle = 0$$

$$\Rightarrow -1 + 9\alpha = 0$$

$$\Rightarrow \alpha = \frac{1}{9}$$

$$\stackrel{5}{=} E_{x+\alpha y} = 4 + 9\alpha^2 - 2\alpha$$

$$\alpha = \frac{1}{9} \quad E_{\min} = 4 + 9\left[\frac{1}{9}\right]^2 - 2\left[\frac{1}{9}\right]$$

$$= \frac{35}{9}$$

Section F

E1: Given $s(t) = r(t+2) - 3r(t+1) + 3r(t) - r(t-1)$

$$\& \ r(t) = tu(t)$$

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$$\stackrel{1}{=} 01 \ r(t+2) = (t+2)u(t+2)$$

$$\stackrel{2}{=} 02 \ r(t+1) = (t+1)u(t+1)$$

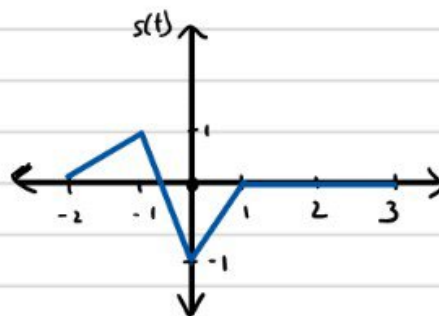
$$\stackrel{3}{=} 03 \ r(t) = tu(t)$$

$$\stackrel{4}{=} 04 \ r(t-1) = (t-1)u(t-1)$$

$$s(t) = (t+2)u(t+2) - 3(t+1)u(t+1) + 3tu(t) - (t-1)u(t-1)$$

Now

$$\stackrel{02}{=} s(t) = \begin{cases} 0 & t < -2 \\ t+2 & -2 \leq t < -1 \\ -2t-1 & -1 \leq t < 0 \\ t-1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases}$$



$$\stackrel{02}{=} \frac{d}{dt} [tu(t)] = u(t)$$

$$\frac{ds}{dt} = u(t+2) - 3u(t+1) + 3u(t) - u(t-1)$$

$$\stackrel{03}{=} \frac{d}{dt} u(t) = \delta(t)$$

$$\frac{d^2 s}{dt^2} = \delta(t+2) - 3\delta(t+1) + 3\delta(t) - \delta(t-1)$$

$$\stackrel{04}{=} \frac{d^3 s}{dt^3} \text{ (third derivative)}$$

$$\frac{d^3 s}{dt^3} = \delta'(t+2) - 3\delta'(t+1) + 3\delta'(t) - \delta'(t-1)$$

E2: Given

$$x(t) = \begin{cases} 0 & |t| \geq 3 \\ t+3 & -3 < t < -1 \\ 2 & -1 \leq t \leq 1 \\ 3-t & 1 < t < 3 \end{cases}$$

01 w.k.T

$$r(t) = tu(t)$$

$$\frac{d}{dt} r(t) = u(t)$$

$$\frac{d}{dt} u(t) = \delta(t)$$

$$\stackrel{02}{=} x(t) = r(t+3) - r(t+1) - r(t-1) + r(t-3)$$

$$\stackrel{03}{=} x(t) = t+3[u(t+3) - u(t+1)] + 2[u(t+1) - u(t-1)] + 3-t[u(t-1) - u(t-3)]$$

$$\begin{aligned} 03 \quad x(t) &= r(t+3) - r(t+1) - r(t-1) + r(t-3) \\ x'(t) &= u(t+3) - u(t+1) - u(t-1) + u(t-3) \\ x''(t) &= \delta(t+3) - \delta(t+1) - \delta(t-1) + \delta(t-3) \end{aligned}$$

E3: Given

$$x(t) = \text{rect}\left[\frac{t-4}{4}\right] + \text{rect}\left[\frac{t+1}{2}\right] - 2\text{rect}(t-1)$$

or We know that

$$\text{rect}\left[\frac{t-a}{T}\right] = 1 ; a - \frac{T}{2} < t < a + \frac{T}{2}$$

$$\text{or } \text{rect}\left[\frac{t-1}{4}\right] = 1 ; -1 < t < 3$$

$$\text{or } \text{rect}\left[\frac{t+1}{2}\right] = 1 ; -2 < t < 0$$

$$\text{or } \text{rect}(t-1) = 1 ; \frac{1}{2} < t < \frac{3}{2}$$

$$02 \quad t < -2 \quad x(t) = 0$$

$$-2 < t < -1 \quad x(t) = 1 \quad (2^{\text{nd}})$$

$$-1 < t < 0 \quad x(t) = 1+1 = 2 \quad (1^{\text{st}} \& 2^{\text{nd}})$$

$$0 < t < \frac{1}{2} \quad x(t) = 1 \quad (1^{\text{st}})$$

$$\frac{1}{2} < t < \frac{3}{2} \quad x(t) = 1-2 = -1 \quad (1^{\text{st}} \& 3^{\text{rd}})$$

$$\frac{3}{2} < t < 3 \quad x(t) = 1 \quad (1^{\text{st}})$$

$$t > 3 \quad x(t) = 0$$

$$x(t) = \begin{cases} 0 & t < -2 \\ 1 & -2 < t < -1 \\ 2 & -1 < t < 0 \\ 1 & 0 < t < \frac{1}{2} \\ -1 & \frac{1}{2} < t < \frac{3}{2} \\ 1 & \frac{3}{2} < t < 3 \\ 0 & t > 3 \end{cases}$$

$$03. \quad E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= 1^2 + 2^2(1) + 1^2\left(\frac{1}{2}\right) + (-1)^2(1) + (1)^2\left(\frac{3}{2}\right)$$

$$E_x = 8$$

$$04 \quad \langle p \rangle = (-2 < t < 3)$$

$$P_x = 0 \quad \text{since } E_x = \text{finite}$$

$$05 \quad x_e(t) = \frac{1}{2} [\text{rect}\left[\frac{t-1}{4}\right] + \text{rect}\left[\frac{t+1}{2}\right] - 2\text{rect}(t-1)]$$

$$+ \text{rect}\left[\frac{t+1}{4}\right] + \text{rect}\left[\frac{t-1}{2}\right] - 2\text{rect}(t+1)$$

$$x_o(t) = \frac{1}{2} [\text{rect}\left[\frac{t-1}{4}\right] + \text{rect}\left[\frac{t+1}{2}\right] - 2\text{rect}(t-1) - \text{rect}\left[\frac{t+1}{4}\right] - \text{rect}\left[\frac{t-1}{2}\right] + 2\text{rect}(t+1)]$$

E4: Given

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$$I_1 = \int_{-\infty}^{\infty} (t^2 - 3t + 1) \delta(t-2) dt$$

$$I_2 = \int_{-\infty}^{\infty} x(t) \delta(2t+4) dt$$

$$I_3 = \int_{-\infty}^{\infty} \delta(t-1) u(2-t) dt$$

$$x(t) = e^{-t} u(t+2) + 2u(-t)$$

$$01. \quad I_1 = -1$$

$$02 \quad I_2 = \int_{-\infty}^{\infty} x(t) \delta(2t+4) dt$$

$$\delta(2t+4) \Rightarrow 2t+4 = x(t+2)$$

$$\delta(at) = \frac{1}{|a|} \delta(t)$$

$$\text{So } \delta(2(t+2)) = \frac{1}{2} \delta(t+2)$$

$$\Rightarrow \int_{-\infty}^{\infty} x(t) \frac{1}{2} \delta(t+2) dt$$

$$I_2 = \frac{1}{2} \int_{-\infty}^{\infty} x(t) \delta(t+2) dt$$

$$I_2 = \frac{1}{2} x(-2)$$

$$\Rightarrow x(-2) = e^{-(-2)} u(-2) + 2u(2)$$

$$= e^2 u(-2) + 2u(2)$$

$$x(-2) = 2$$

$$\Rightarrow I_2 = \frac{1}{2}(2) = 1$$

$$03 \quad I_3 = \int_{-\infty}^{\infty} \delta(t-1) u(2-t) dt$$

$$\int f(t) \delta(t-a) dt = f(a)$$

$$\Rightarrow I_3 = 1$$

04 used formulae

$$01 \quad \delta(at+b) = \frac{1}{|a|} \delta\left(t+\frac{b}{a}\right)$$

$$02 \quad \delta(2t+4) = \frac{1}{2} \delta(t+2)$$

$$03 \quad \int f(t) \delta(t-a) dt = f(a)$$

05 Hence

$$I_1 = -1$$

$$I_2 = 1$$

$$I_3 = 1$$